Summary cards

9 cards. Read one card at a time, then try to say it in your own words.

1 What is a magnet?



Bar magnet *U-shaped magnet* *Ring magnet*

- A magnet pulls some things, like iron pins, towards itself.
- Long ago, people found magnets in nature. These are called lodestones.
- Today most magnets are made by people. They are called artificial magnets.
- Magnets come in many shapes: bar magnet, U-shaped magnet and ring magnet.

Lodestone

A magnet found in nature.

Artificial magnet

A magnet made by people.

EXAMPLE

A pencil box, a purse and a duster for the writing board can have a magnet in them.

Read in the chapter: Introduction

2 Magnetic and non-magnetic materials

- Magnetic materials are attracted towards a magnet. They stick to it.
- Iron, nickel and cobalt are magnetic materials.
- Non-magnetic materials are not attracted towards a magnet.
- Wood, plastic, glass and cardboard are non-magnetic materials.

Attracted

Pulled towards something.

EXAMPLE

Bring a magnet near a pencil. The wood does not stick, so wood is non-magnetic.

REMEMBER

To know if a material is magnetic, test it with a magnet. Do not guess.

Read in the chapter: 4.1 Magnetic and Non-magnetic Materials

3 The two poles of a magnet



- Iron filings stick the most near the two ends of a bar magnet.
- These two ends are called the poles of the magnet.
- Every magnet has two poles, whatever its shape.
- Poles always come in pairs. Break a magnet and each piece still has two poles.
- A magnet with only one pole cannot exist.

Iron filings

Very small pieces of iron.

North pole of a magnet

The end that points towards the north when the magnet hangs freely.

South pole of a magnet

The end that points towards the south when the magnet hangs freely.

REMEMBER

Very few iron filings stick at the middle of a bar magnet.

Read in the chapter: 4.2 Poles of Magnet

4 A hanging magnet shows north and south



- Hang a bar magnet by a thread so that it can turn freely.
- It always comes to rest along the north-south direction.
- The end pointing north is the North pole. The other end is the South pole.
- This happens because our Earth behaves like a giant magnet.

Freely suspended

Hanging so that it can turn easily in any direction.

EXAMPLE

Hang an iron bar the same way. It can rest along any direction, so it is not a magnet.

REMEMBER

Only a magnet rests along north-south every time. This is a test for a magnet.

Read in the chapter: 4.3 Finding Directions

5 The magnetic compass



- A magnetic compass is a small box used to find directions.
- Inside it is a magnet shaped like a needle. The needle can rotate freely.
- The needle always rests along the north-south direction.
- The end of the needle that points north is usually painted red.
- Turn the box until north and south on the dial line up with the needle.

Dial

The round plate under the needle with the directions marked on it.

Matsya-yantra

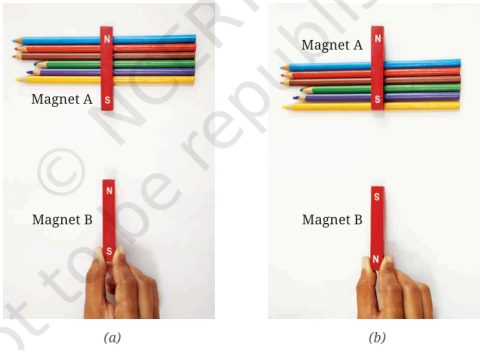
An old Indian compass. It was a magnetised fish-shaped iron piece kept in a vessel of oil.

EXAMPLE

Make your own compass. Move one pole of a magnet along an iron needle 30 to 40 times, always from the same end. Pass the needle through a cork and float it in water.

Read in the chapter: 4.3 Finding Directions

6 Attraction and repulsion



(a) (b)

- Unlike poles (North and South) attract each other. This pull is called attraction.
- Like poles (North-North or South-South) repel each other. This push is called repulsion.
- A magnet attracts both ends of an iron bar. It never repels the iron bar.
- So repulsion is the sure way to tell that an object is a magnet.
- A compass needle is also a magnet. A like pole brought near pushes its end away.

Repel

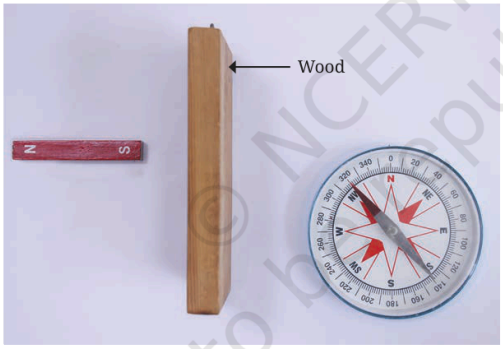
To push away.

REMEMBER

Like poles repel. Unlike poles attract.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

7 Magnets work through some materials



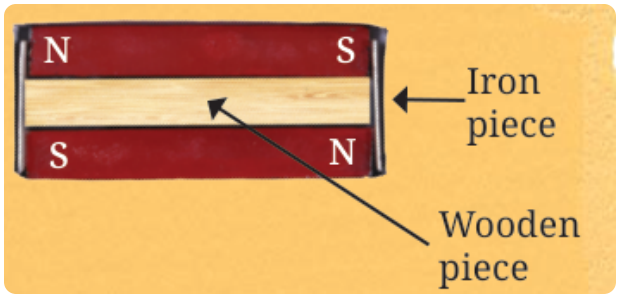
- A magnet can move a compass needle without touching it.
- Put wood, cardboard, plastic or glass in between. The needle still moves the same way.
- So the magnetic effect can act through non-magnetic materials.

EXAMPLE

A magnet held outside a glass of water can pull out a steel paper clip. Your fingers stay dry.

Read in the chapter: 4.4 Attraction and Repulsion between Magnets

8 Fun with magnets and taking care of them



- Fun ideas: a garland of pins, steel balls moved through a maze, matchbox cars with magnets.
- Do not heat, drop or hammer a magnet.
- Keep magnets away from mobile phones and remote controls.
- Store bar magnets in pairs, with unlike poles on the same side and wood in between.
- Place a piece of soft iron across each end of the pair.

REMEMBER

Poles can be marked in different ways. N and S, a white dot for North, or red for North and blue for South.

Read in the chapter: 4.5 Fun with Magnets

9 Working like a scientist

- Each activity in this chapter asks you to do one of these actions.
- First say what you think will happen. Then try it and watch carefully.
- Write down what you see. Then decide what it tells you.

Predict

Say what you think will happen before you try.

Observe

Look carefully at what happens.

Record

Write down what you see, for example in a table.

Explore

Try things out to find something new.

Investigate

Look into something step by step to find the answer.

Experiment

A test you do to check an idea.

Construct

Make or build something, like your own compass.

Conclude

Decide what your observations tell you.

> **REMEMBER**
> A conclusion must come from what you observed, not from a guess.

Read in the chapter: 4.1 Magnetic and Non-magnetic Materials